Correction: Uniform Vorticity Envelope versus Cubic Differential Hypothesis

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Abstract

This addendum separates the uniform-envelope assumption in the checked Lean reduction from a genuinely differential assumption on the actual vorticity supremum. It records the conditional comparison proved in the accompanying source patch, tests a proposed coercivity mechanism on explicit periodic solutions, and identifies the remaining analytic obligation. Version 1.1 adds a Lean proof that the decaying periodic shear is an actual solution in the repository's solution structure, with its exact vorticity supremum, so the data-dependent cubic bound and the fixed-constant obstruction now hold for an actual solution, not only for a scalar profile. It is not a proof of general Navier–Stokes regularity. Earlier published PDFs and archives are unchanged.

1 The corrected distinction

The baseline source is `BenFrohman/NS_Millennium_Proof` at commit `c98fe859867affeba4508b3e51583cfa6b20087c`. Historically `DirectionCurvatureBound` abbreviates `PeriodicDirectionDepletionBound`. Its actual body assumes that, for every smooth divergence-free periodic datum, there are positive constants C, κ such that every represented local solution satisfies

$$M(t) \le \max\{M(0), C/\kappa\}, \qquad 0 \le t < T.$$

Here $M(t) = \|\operatorname{curl} u(t)\|_\infty$. The constants may depend on viscosity and the datum, but not on the solution or its lifespan. This definition does not assert a differential inequality. The accurate public name is now `UniformVorticityEnvelopeBound`; the original body, historical aliases and original conditional theorem are preserved.

The separate `CubicVorticityDifferentialBound`(u, τ, C, κ) explicitly requires $\tau \ge 0$, $C, \kappa > 0$, bounded spatial vorticity-norm ranges, ordinary differentiability of the actual M at every point of $[0, \tau]$, and

$$M'(t) \le CM(t)^2 - \kappa M(t)^3 \quad (0 \le t \le \tau).$$

The boundedness guard matters for the real-valued conditional supremum used in Lean. No differential estimate or differentiability of a supremum is inferred merely from smoothness of the velocity.

2 The checked conditional consequences

The existing `comparison_ode_ceiling`, with all its hypotheses, gives the compact envelope above. The new source proves this for the actual curl supremum, then bounds the actual curl pointwise using spatial boundedness. On an actual `IsPeriodicSmoothSolutionOn`, compatible compact hypotheses for every $0 \le \tau < T$ with the *same* coefficients give the envelope on $[0, T)$; the initial identity comes from that solution's definition. A zero-trajectory regression shows that the differential predicate is inhabited.

`clay_reduces_to_uniform_vorticity_envelope` accurately names the existing conditional reduction: the uniform envelope for every positive viscosity, *and* the separately assumed periodic BKM continuation principle, imply `Clay_B_exact`. Neither premise is established by that reduction.

The Stage6 additions check the derivative of $M_0 e^{-at}$, its cubic estimate with $C = \kappa M_0$, and the fixed-coefficient obstruction below. These are *scalar* calculus and algebra theorems.

New in v1.1: an actual solution. The file is `NS_Millennium_Proof/Modules/ShearSolution.lean`. For the single shear $v_A(t,x) = Ae^{-\nu k^2 t}\sin(kx_3)\,e_1$, $k = 2\pi$, $p = 0$ (Section 3.2), Lean proves against the repository's own definitions of `curl`, `laplacian`, `convective`, `div`, `vorticity_sup_norm` and `IsPeriodicSmoothSolutionOn`:

- `shear_isPeriodicSmoothSolutionOn`: for all A, ν, T, v_A with $p = 0$ is a jointly C^∞, $\mathbb{Z}^3$-periodic, divergence-free solution on $[0, T)$ with initial value $v_A(0)$. The proof computes $\partial_t v_A = -\nu k^2 v_A$ (two-sided, which gives the one-sided derivative within $[0, \infty)$ used by the predicate), $(v_A \cdot \nabla)v_A = 0$ because v_A has no x_3-component, $\operatorname{div} v_A = 0$ because v_A depends only on x_3, and $\Delta v_A = -k^2 v_A$, so $\partial_t v_A + (v_A \cdot \nabla)v_A + \nabla p = \nu \Delta v_A$.
- `shear_vorticity_sup_norm`: for $A > 0$, $\sup_x |\operatorname{curl} v_A(t,x)| = kAe^{-\nu k^2 t}$. Indeed $\operatorname{curl} v_A = (0,\, kAe^{-\nu k^2 t}\cos(kx_3),\, 0)$; its norm is at most $kAe^{-\nu k^2 t}$ and equals it at $x = 0$.
- `shear_cubic_bound_data_dependent`: for $A > 0$, $\nu \ge 0$, $\kappa > 0$, $\tau \ge 0$, the actual v_A satisfies `CubicVorticityDifferentialBound` with the data-dependent $C = \kappa \cdot kA$.
- `shear_fixed_cubic_constants_fail`: for fixed $C \ge 0$, $\kappa > 0$ and $\nu \ge 0$ there is $A > 0$ such that v_A is an actual solution and violates the cubic bound at $t = 0$ (amplitude $A = M/k$ with M from `fixed_cubic_constants_fail_for_some_amplitude`).
- `shear_envelope_nonvacuous`: for $A > 0$, $\nu \ge 0$, $\kappa > 0$, $v_A(0) \ne 0$ and the envelope conclusion $\sup|\omega(t)| \le \max(\sup|\omega(0)|, C/\kappa)$ with $C = \kappa kA$ holds for all $t \in [0, T)$, obtained from the conditional envelope theorem.

The predicate asks for ordinary two-sided differentiability of M; for v_A, defined by the same formula for all real t, $M(t) = kAe^{-\nu k^2 t}$ is smooth. Still not in Lean: the Beltrami family below (a hand calculation), the direction field ξ and the statement $|\nabla \xi|^2 = 0$ for the shear, and any estimate for general solutions.

3 What the PDE identity supplies

For a classical unforced incompressible solution, let $\omega = \operatorname{curl} u$, $S = (\nabla u + (\nabla u)^T)/2$ and $\rho = |\omega|$. Then

$$(\partial_t + u \cdot \nabla)\frac{|\omega|^2}{2} = \omega \cdot S\omega + \nu \Delta \frac{|\omega|^2}{2} - \nu |\nabla \omega|^2.$$

Where $\rho > 0$, write $\omega = \rho\xi$, $|\xi| = 1$. Differentiating gives

$$|\nabla\omega|^2 = |\nabla\rho|^2 + \rho^2|\nabla\xi|^2,$$
$$(\partial_t + u\cdot\nabla - \nu\Delta)\rho = \rho\,\xi\cdot S\xi - \nu\rho|\nabla\xi|^2.$$

At a positive spatial maximum $\nabla\rho = 0$ and $\Delta\rho \le 0$. These facts do not give a lower bound on direction curvature. A general maximum-in-time argument also needs a justified one-sided/Dini derivative treatment of maximizing-branch switches, not an automatic ordinary derivative of M.

3.1 Beltrami family: fixed coefficients versus data dependence

On the unit torus take $\nu > 0$, $t \ge 0$, $k = 2\pi$, $a = \nu k^2$, $A > 0$, and

$$u_A(t,x) = Ae^{-at}(\sin(kx_3), \cos(kx_3), 0), \qquad p = 0.$$

Direct differentiation gives $\operatorname{div} u_A = 0$, $(u_A\cdot\nabla)u_A = 0$, $\Delta u_A = -k^2 u_A$, $\partial_t u_A = \nu\Delta u_A$ and $\operatorname{curl} u_A = k u_A$. Consequently

$$M_A(t) = kAe^{-at}, \qquad M_A' = -aM_A.$$

At zero, the cubic estimate is equivalent to $\kappa M_A(0)^2 - CM_A(0) - a \le 0$. For fixed $C \ge 0$, $\kappa > 0$ and $a \ge 0$, this fails at sufficiently large amplitude. The Lean theorem `fixed_cubic_constants_fail_for_some_amplitude` proves the quantified scalar obstruction. This does *not* refute the data-dependent envelope: choosing $C = \kappa M_A(0)$ makes the cubic right-hand side $\kappa M_A(t)^2(M_A(0) - M_A(t)) \ge 0$, while $M_A' \le 0$. The latter special-case sufficient estimate is also checked in Lean for the scalar profile.

3.2 Single shear: direction-only coercivity fails

Instead take the actual periodic heat shear

$$v_A(t,x) = Ae^{-at}(\sin(kx_3), 0, 0), \qquad p = 0.$$

The same divergence, convection and heat calculation verifies the unforced Navier–Stokes equations. Here

$$\omega_A(t,x) = (0, kAe^{-at}\cos(kx_3), 0).$$

At $x_3 = 0$, its norm attains the positive global maximum $M = kAe^{-at}$. In a neighborhood $\cos(kx_3) > 0$, so $\xi = (0,1,0)$ is constant and $|\nabla\xi|^2 = 0$. Thus the bare premise

$$\kappa M^2 \le \nu|\nabla\xi|^2$$

from `viscous_direction_dissipation_bridge` fails for *every* $\kappa > 0$, even if it depends on this datum. This refutes promotion of that premise to an unconditional assertion at every positive maximum, not the existing implication which assumes it. Nor does it refute a different estimate restricted above a data-dependent threshold. The actual damping here comes from amplitude curvature:

$$\omega_A\cdot\Delta\omega_A = -k^2M^2, \qquad M' = -\nu k^2 M.$$

Discarding amplitude curvature loses this real viscous contribution.

3.3 Scaling is not a coercivity proof

For the parabolic rescaling $u_\lambda(t,x) = \lambda u(\lambda^2 t, \lambda x)$, one has $M_\lambda(t) = \lambda^2 M(\lambda^2 t)$ and the time derivative scales by λ^4 wherever it exists. A cubic inequality therefore transforms with $C_\lambda = C$ and $\kappa_\lambda = \kappa/\lambda^2$, not fixed positive κ. Likewise $\nu|\nabla\xi|^2$ scales by λ^2 whereas M^2 scales by λ^4. This does not contradict data-dependent coefficients. On the fixed unit torus the displayed transformation preserves unit periodicity for positive integer λ; arbitrary domain rescalings are not the same torus. The separate scale-homogeneous premise $\kappa_{\text{eff}} M \le \nu|\nabla\xi|^2$ in the source only yields $M' \le (C - \kappa_{\text{eff}})M^2$, not cubic damping in M. Scaling supplies neither coercivity premise.

4 Endpoint representation and remaining obligation

The Lean local-solution structure constrains only nonnegative times. Extend the zero forward solution by $(\sin(kx_3), \cos(kx_3), 0)$ for negative times. It is still the zero solution on $[0,T)$, but its actual vorticity supremum jumps from k to zero at time zero. Hence ordinary two-sided differentiability of M at zero cannot follow for every represented solution. This is not an obstruction to right-sided PDE smoothness. The globally zero trajectory, unlike this other extension, passes the new regression. The compact differential predicate is an explicit additional condition on a given trajectory, not a universal PDE theorem.

A sufficient net-source estimate at the relevant positive maxima would be

$$\omega \cdot S\omega + \nu\omega \cdot \Delta\omega \le CM^3 - \kappa M^4,$$

with correctly quantified positive coefficients uniform in time and lifespan. Division by M then gives the cubic rate after a justified maximum-derivative argument. The repository proves the algebra conditionally, not this general source estimate. An endpoint L^∞ Calderon–Zygmund bound cannot silently supply it. A suitable source estimate and a one-sided/Dini comparison bridge remain analytic/formal obligations. No general regularity breakthrough is claimed.

5 Reproduction and evidence boundary

Lean source: `https://github.com/BenFrohman/NS_Millennium_Proof`, code commit `1bcbb3845f99543b1b77d07292188ce13eb74315` (pull request #26). Lean `leanprover/lean4:v4.30.0-rc1`; Mathlib tag `v4.30.0-rc1`, locked commit `0c154d67103f74be3a0f2c509f72ccbf5be9f2a7`; all pins in `lake-manifest.json`. One command from the repository root:

```
lake exe cache get && lake build && python3 scripts/check_envelope_axioms.py
```

GitHub Actions ran this in `https://github.com/BenFrohman/NS_Millennium_Proof/actions/runs/42925945732` (passed). The checker fails if any of the 37 audited declarations in `scripts/EnvelopeCorrectionAudit.lean` depends on anything beyond `propext`, `Classical.choice` and `Quot.sound`; its output, grouped by namespace (each printed line was identical apart from the name), was

```
Each of the following 37 declarations printed
  depends on axioms: [propext, Classical.choice, Quot.sound]
FrohmanianTether.HighSetDepletionAndBKM.
    vorticity_envelope_of_cubic_differential_bound
    pointwise_vorticity_envelope_of_cubic_differential_bound
    periodic_solution_envelope_of_compact_cubic_bounds
    cubicVorticityDifferentialBound_zero
```

```
    clay_reduces_to_uniform_vorticity_envelope
NS_Millennium_Proof.Stage6.
    exponential_profile_hasDerivAt exponential_profile_satisfies_cubic_bound
    fixed_cubic_constants_fail_for_some_amplitude
NS_Millennium_Proof.ShearSolution.
    K_pos e0_apply single_apply_two hasFDerivAt_F fderiv_F_apply
    hasFDerivAt_F_coord fderiv_F_coord_apply convective_F div_F laplacian_F
    curl_F sin_profile_deriv cos_profile_deriv shearU_eq shearU_zero
    shear_time_deriv gradient_zero latticeVec_two shear_periodic
    shear_contDiff shear_isPeriodicSmoothSolutionOn shear_vorticity
    shear_vorticity_norm shear_vorticity_bdd shear_vorticity_sup_norm
    shear_M_eq shear_cubic_bound_data_dependent
    shear_fixed_cubic_constants_fail shear_envelope_nonvacuous
PASS: 37 declarations, standard axioms only
```

The same run's whole-library inventory (`scripts/CertificateAxiomInventory.lean`, which aborts on any non-standard axiom) printed `TOTAL_THEOREM_DECLARATIONS530`. No new axiom, `sorry` or `native_decide` is introduced. Standard-axiom checks establish neither a missing analytic hypothesis nor novelty, and a kernel check is not a statement about general Navier–Stokes solutions.

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